NACA Airfoil & Boundary Layer Simulator

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Application URL: `https://poompm.github.io/Airfoil-simulator-vibe/`

1 Overview and Description

The Advanced NACA Aerodynamics & Boundary Layer Simulator is a fully interactive, web-based engineering tool designed to model external incompressible viscous flow over standard NACA 4-digit airfoils. The underlying physics and governing principles of this simulation are directly based on the external incompressible viscous flow course materials and textbook references provided (specifically drawing from principles of boundary layer concepts, drag and lift coefficients, pressure gradients, and separation phenomena). Built using HTML5 Canvas and advanced vector field mathematics, the application bridges theoretical fluid mechanics with real-time visual feedback. It simulates complex aerodynamic phenomena including laminar-to-turbulent boundary layer behavior, adverse pressure gradients, surface flow tangency, and aerodynamic stall.

2 Core Features & Technical Capabilities

- **Dynamic NACA 4-Digit Generation:** Users can customize airfoil profiles in real time by adjusting the Maximum Camber (M), Camber Position (P), and Thickness (T), instantly rendering exact geometry using standard aeronautical equations.
- **Rigorous Fluid Physics Engine:**
  - **No-Slip Boundary Condition:** Simulates fluid velocity decay near the solid surface.
  - **Surface Tangency:** Streamlines mathematically align with local camber slopes to naturally hug the airfoil contour.
  - **Separation and Wake Modeling:** Automatically computes the separation point ($\partial u/\partial y|_{y=0} = 0$) and models chaotic eddies within the low-pressure wake zone during a stall.
- **Dual Visualization Modes:** Switch seamlessly between Streamline Particles (motion-blurred moving tracer particles) and Continuous Lines (steady-state vector pathlines).
- **Live Telemetry & Metrics:** Real-time computation of the Lift Coefficient (C_L), Drag Coefficient (C_D), and flight status indicators (Attached Flow vs. Stall).
- **Dynamic Infographic Annotations:** On-screen visual indicators automatically track and point to critical aerodynamic regions, including the Stagnation Point, Viscous Boundary Layer, Separation Point, and Turbulent Wake.

3 User Guide & Instructions

3.1 Step-by-Step Operation

1. **Access the Application:** Open the published web link in any modern web browser: `https://poompm.github.io/Airfoil-simulator-vibe/`
2. **Select Visualization Mode:** Use the dropdown menu at the top of the control panel to choose between Streamline Particles (best for viewing dynamic flow momentum) or Continuous Lines (best for steady-state field patterns).
3. **Configure Airfoil Geometry:**

- **Max Camber (M, %):** Adjusts the first digit (maximum camber as a percentage of chord).
- **Camber Position (P, %):** Adjusts the second digit (location of max camber along the chord).
- **Thickness (T, %):** Adjusts the final two digits (maximum thickness profile as a percentage of chord).

4. **Adjust Flight Conditions:**
   - **Angle of Attack (α):** Use the slider to pitch the airfoil nose up or down relative to the relative wind (α is measured positive counterclockwise/clockwise relative to the horizontal wind vector). Push past $\sim 13.5°$ to trigger an aerodynamic stall.
   - **Relative Wind Speed:** Adjusts the free-stream velocity magnitude (U_∞) of the flow field.

5. **Analyze Telemetry & Infographics:**
   - Observe how the red separation point indicator moves forward along the upper surface as the Angle of Attack increases.
   - Monitor live changes in C_L and C_D within the metrics panel.

4 Evidence of AI Usage in Development

AI was utilized as an end-to-end pair programming and technical assistant throughout the development of this interactive web application. The complete record and chat transcript demonstrating how AI was used can be accessed via the link below:

- **AI Collaboration Session URL:** `https://share.gemini.google/YlogoWty9HlB`
- **Scope of AI Assistance:**
  - *Physics Formulation:* Translating fluid mechanic equations from course materials (boundary layer velocity decay, surface tangency, and adverse pressure gradients) into programmatic vector math algorithms.
  - *Code Architecture:* Structuring the HTML5 Canvas loop, responsive layout grids, and real-time UI binding controls.
  - *Debugging & Iteration:* Refining the coordinate systems, sign conventions for Angle of Attack (α), and optimizing rendering frames.